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PALAEOMAGNETISM OF THE CAMBRIAN PURPLE SANDSTONE FROM THE SALT RANGE, WEST PAKISTAN

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Samples from two localities near Khewra have been examined palaeomagnetically. At one locality the samples are shown to be unstable, but at the other they respond to thermal cleaning and are shown to be extremely stable. In a rare occurrence, a 10 cm core drilled through one sample showed two polarity changes occurred during the time taken for its deposition. Changes in intensity recorded from intermediate directions in the core suggest the field dropped to a quarter or fifth of its normal value during the polarity changes. An approximate rate of sedimentation of between 0.1 and 1 cm per thousand years is estimated for these sediments. The new palaeomagnetic pole for the Indian sub-continent enables the position of this continent to be determined uniquely with respect to the other Gondwanic continents. India is placed alongside East Africa, but not adjacent to Western Australia. The configuration deduced is almost identical to that originally proposed by du Toit.

1. Introduction

It has been shown that the relative positions of two or more continents can be determined uniquely from palaeomagnetic measurements, if they have remained in a fixed position relative to each other during a period in which substantial polar wandering has occurred [1,2]. Results from the Palaeozoic of South America [3] and Africa [4] show that substantial polar movement occurred during some of that period. McElhinny [5] has demonstrated that the polar paths for Africa and South America coincide for virtually the whole of the Palaeozoic when the South Atlantic is closed. The position of Australia with respect to these two continents could only be inferred from Upper Palaeozoic results [6]. It is therefore important in attempting to reconstruct Gondwanaland that Lower Palaeozoic palaeomagnetic measurements be obtained for Australia, India and Antarctica. This paper therefore reports the first results from the Lower Palaeozoic of the Indian sub-continent, as part of a programme presently being undertaken to rectify the situation.

Some confusion has arisen in the past, since there are results reported from two formations in India, which the authors have incorrectly claimed to be of Cambrian age [7,8]. An age of approximately 500 my was cited for the Malani rhyolites of Rajastan, when palaeomagnetic data were first presented [7]. Rb-Sr measurements [9] show that these rocks are 745 ± 10 my old and are therefore quite clearly Precambrian. Also Sahasrabudhe and Mishra [8] have reported results from the Kaimur Series of the Vindhyan System, stating these sediments to be of Middle Cambrian age. Crawford [9] has shown that the base of the Kaimur Series is about 1150 my old from Rb–Sr ages on the Panna diamond pipe, that intrudes it. These rocks are also quite clearly Precambrian in age.

2. Geology

Palaeozoic rocks are not well developed in the Indian sub-continent. They occur in the Himalayas, particularly in the Spiti region [10], but the area is inaccessible and relatively unexplored. It has been suggested that the Bhander Series of the Vindhyan

System is of Lower Palaeozoic age [11], but this appears to be largely speculative, and these rocks could just as likely be late Precambrian. The only remaining area where Lower Palaeozoic rocks are developed is in the Salt Range of West Pakistan (fig. 1), where the succession contains some Lower Cambrian sediments [10]. Upper Palaeozoic rocks are developed at the base of the Gondwana Group, and some results from the Upper Carboniferous Talchir Series have recently been reported [12].

The sequence of the Cambrian rocks in the Salt Range is given in table 1. The Saline Series at the base may well extend back into the Precambrian. Near the top, it contains the strange "Khewra trap", an alkali rock which has been termed khewraite [13]. The exposure is too poor to sample for palaeomagnetic purposes. An extensive study of the succession has been undertaken by Schindewolf and Seilacher [14], whose report has been reviewed in English by Teichert [15]. The Purple Sandstone, a partly purple, partly reddish and yellow-brown sandstone contains undiagnostic anthropod trails and the trace fossil

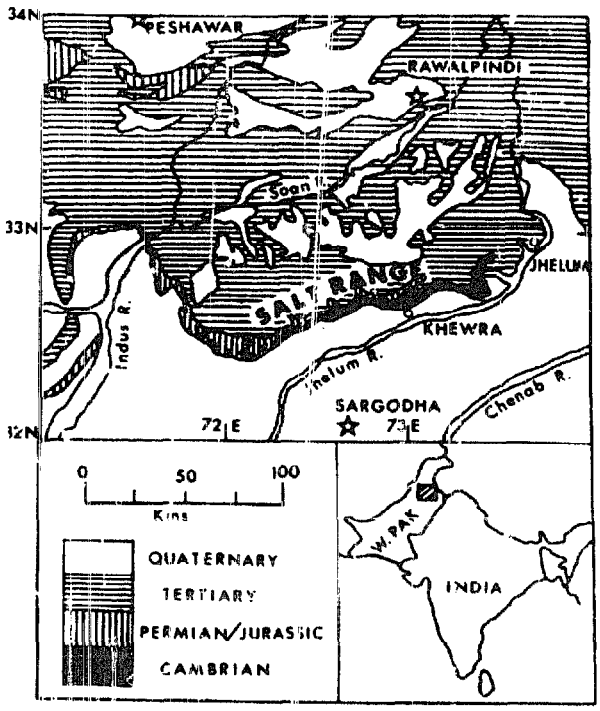


Fig. 1. Geological sketch map of the Salt Range and environs, West Pakistan. Samples were collected near Khewra.

Table 1
Sequence in the Salt Range, West Pakistan.

Upper Carboniferous unconformity	Talchir Boulder-Bed
Lower Cambrian – 300 m	Salt Pseudomorph Beds Magnesium Sandstone (dolomite) Neobulus Shales Purple Sandstone
Late Precambrian – Cambrian	Khewra trap Saline Series – 500 m +

Rusophycus didymus, interpreted as the burrows of small trilobites similar to those occurring in the Magnesium sandstone above.

The Purple Sandstone is conformably overlain by the Neobulus shales and the so-called Magnesium sandstone, which is really a dolomite. Both these units are characterised by the occurrence of *Redlichia noethingi* and *Botsfordia granulata*. They both therefore belong to the *Redlichia* stage and are therefore placed in the upper part of the Lower Cambrian. The fourth unit, the Salt Pseudomorph beds, is conformable with the Magnesium sandstone. Because of the conformable position of the entire sequence and the transitional nature of their boundaries, Schindewolf and Seilacher [14] are inclined to regard all four units as of Early Cambrian age. Of these units only the Purple Sandstone held any promise for palaeomagnetic purposes. It represents the only likely source of palaeomagnetic data for the Lower Palaeozoic of the Indian sub-continent.

3. Sampling and measurement

Oriented hand samples were collected from an exposure in the Khewra Gorge, about 2 km upstream from Khewra (lat. 32.7°N, long. 73.0°E) (fig. 1). The 13 samples cover a stratigraphic thickness of about 50 m, the beds dipping at about 36° towards the north-west at this point. A further 10 samples were collected from an exposure on the Khewra to Choa Saiden Shah road about 2.5 km from Khewra. A vertical thickness of about 30 m was sampled, but at this exposure the beds have only been tilted very slightly and dip 5° towards the north.

From each oriented block both 2.2 cm by 2.2 cm cylinders and 3.5 cm diameter by 1.0 cm thick disks were cored and sliced in the laboratory. Measurements of direction and intensity were carried out on astatic magnetometers [16]. The cylinders and disks have approximately the same volume and the purpose was to investigate whether any systematic differences occur from using these sizes or whether one has any advantage over the other as far as sensitivity was concerned. No differences of any significance were observed nor does there appear to be any difference in apparent sensitivity.

Samples from the Khewra Gorge section showed widely scattered directions upon initial measurement; more often than not, the four specimens measured from each sample did not agree (fig. 2a). Intensities were weak, the mean being 2.5×10^{-6} emu cm^{-3}, with the 52 values ranging from 0.4 to 5.3×10^{-6} emu cm^{-3}. By contrast the NRM of samples from the road section were quite well grouped, taking into account that both normal and reversed directions are obviously represented (fig. 2b). Intensities were noticeably higher, the 40 specimen values ranging from 0.7 to 26.4×10^{-6} emu cm^{-3} with a mean of 9.6×10^{-6} emu cm^{-3}.

4. Thermal demagnetization

Selected specimens from each sample were subjected to the usual thermal demagnetization procedure involving heating to successively higher temperatures and cooling in field free space. The furnace is at the centre of an automatically controlled system of coils cancelling the geomagnetic field. For the vertical and north-south components, four-coil systems are used [17], and a Helmholtz pair cancels the small residual east-west component. The feed-back system keeps the residual field at the specimens to less than 0.5 gamma during the course of a heating and cooling run [18].

Typical thermal demagnetization curves are shown in fig. 3. Specimens from the Khewra Gorge section exhibited rather odd behaviour. After initial decrease in intensity up to 300°C, an increase was usually observed at 400°C before it decreased again up to the Curie Point (fig. 3a). This increase in intensity was associated with a sudden increase in susceptibility observed from the measurements on the astatic magnetometer. Also there appeared to be a tendency for the samples to acquire a temporary component of magnetization on being removed from the field free space at room temperature. Since the magnetization

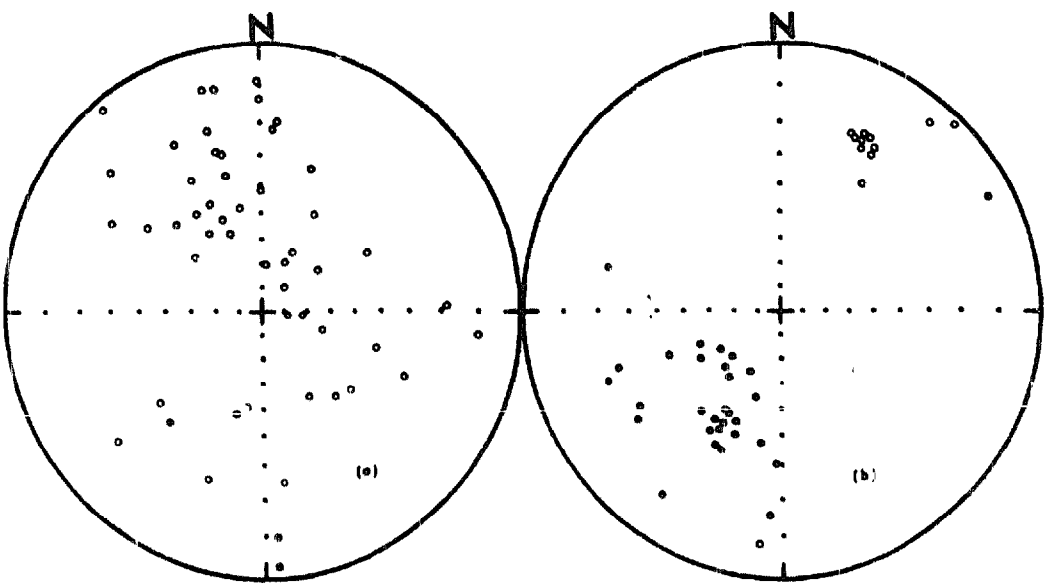


Fig. 2. Stereographic projection of NRM of samples from (a) Khewra Gorge, and (b) Khewra to Choa Saiden Shah road. Solid symbols, lower hemisphere; open symbols, upper hemisphere. Star is the axial dipole field.

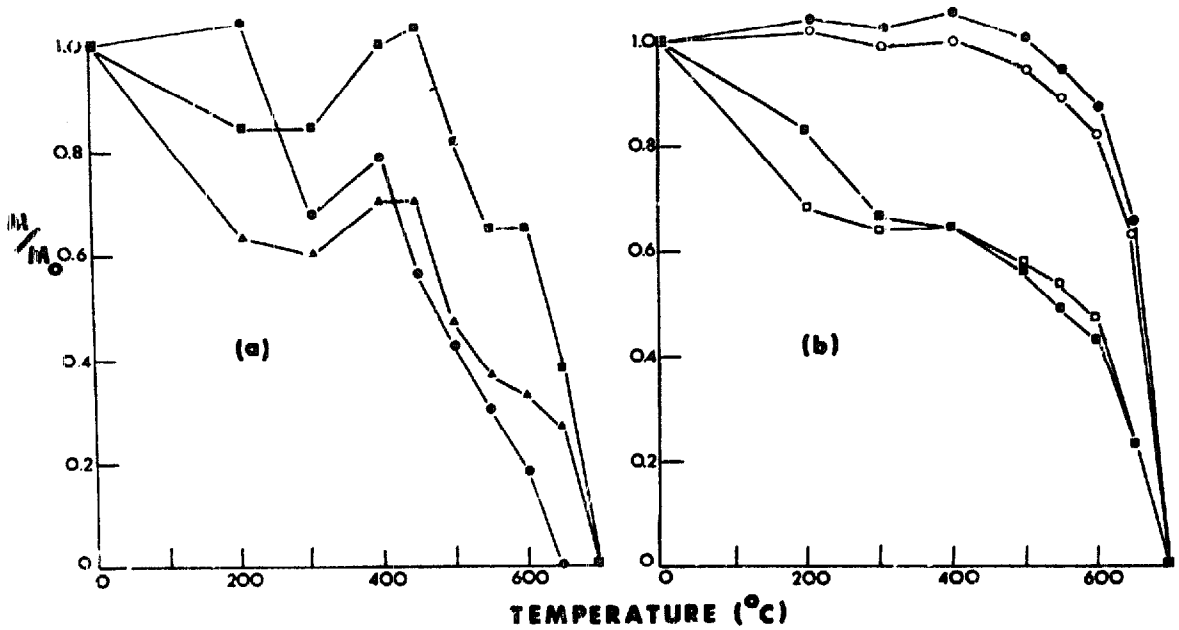


Fig. 3. Thermal demagnetization curves. (a) Unstable samples from the Khewra Gorge, and (b) stable samples from the road section.

Table 2
Directions of magnetization of samples from the Khewra road section after thermal cleaning at 600°C. The site location is lat. 32.7°N, long. 73.0°E.

		Mean direction				Pole position		
Sample	*N*	*D*	*I*	*R*	*k*	Lat. S	Long. E	
C14	3	208½	47½	2.856	13.9	23	46	
C15	3	209½	58	2.897	19.5	14	50	
C16	7	206½	22½	6.948	116	39	39	
	6	40	−21	5.834	30.1	32	24½	
C17	3	192	27	2.883	17.2	41½	57½	
C18	3	211	49½	2.905	21.1	20½	44½	
C19	3	240	13½	2.992	254	20½	6	
C20	3	26	−23½	2.999	2270	38½	40	
C21	3	32	−24½	2.997	748	35½	34	
C22	5	247½	17	5.534	8.58	13½	3	
C23	3	238	29½	2.979	95.7	16½	14½	
Overall mean	11	218	31½	10.334	15.0			(α_{95} = 12°)
Mean of poles	11	–	–	10.426	17.4	28	32	(A_{95} = 11°)

N = no. of specimens, *D* = declination, *I* = inclination, *R* = resultant of *N* unit vectors, *k* = precision parameter. α_{95} and A_{95} are the radii of the circles of 95% confidence about the mean direction and mean pole respectively.

was very weak and specimens had to be measured fairly close to the lower magnet, this also tended to impart temporary components during the course of measurement. No systematic changes in direction were observed and in most cases there was little agreement between specimens from the same sample. The magnetization of all these samples appears to be quite unstable.

The behaviour of specimens from samples from the road section was quite different. These samples appear to have extremely stable magnetization. Their thermal demagnetization curves (fig. 3b) are of the usual two types, thermally distributed (4 samples) and thermally discrete (6 samples) [19]. Small changes in direction were observed on heating in most cases, and all specimens were then thermally cleaned at 600°C. The mean direction for each of the samples after cleaning is given in table 2, together with the overall mean. In one sample both normal and reversed directions were observed and these have been treated as separate vectors. This is discussed in more detail below. The overall mean, taken irrespective of sign, has therefore been calculated from 11 vectors, which are shown in fig. 4a.

5. Reversals and the rate of sedimentation

It was noticed that the specimens originally cored from one sample (C16) showed both normal and reversed magnetizations. At first it was thought an error might have been made in marking some of the cores from this sample, but this was found not to be the case. It was noted, however, that the cores from this sample had been drilled parallel to the bedding plane and that the two polarities could be correlated with cores which effectively came from different horizons. To investigate this further, the sample was cored in a direction at right angles to the bedding plane. A 3.5 cm diameter core about 10 cm long was thus obtained and was then sliced into 11 disks, each approximately 0.7 cm thick. These 11 disks then represent a time sequence through this sample. After intial measurement each sample was thermally cleaned at 600°C.

Succesive directions of magnetization through the core are illustrated in fig. 4b, and also in fig. 5 where the variation in intensity is also shown. Two polarity changes are observed. Three disks at the bottom of the core are reversely magnetized, followed by two

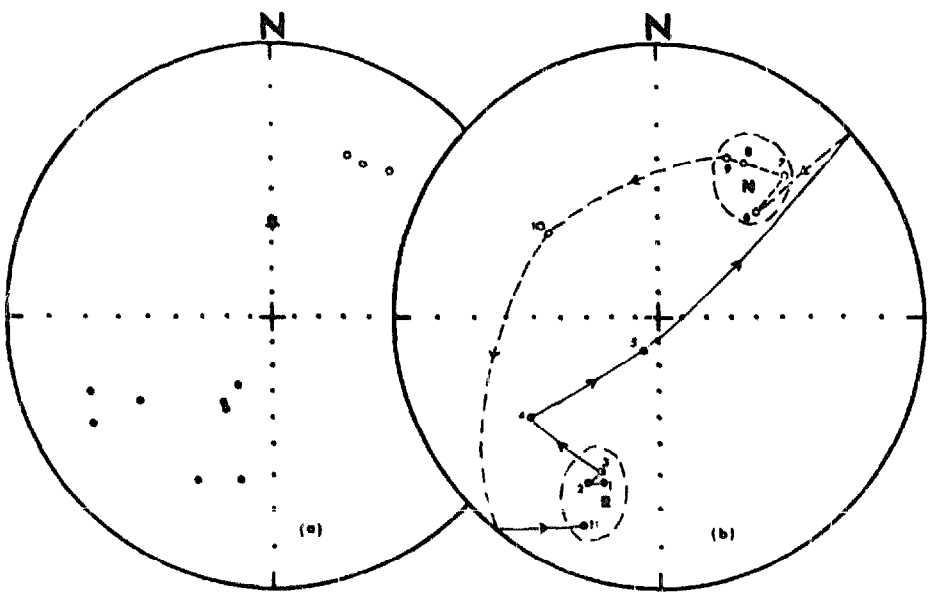


Fig. 4. Stereographic projection of directions of magnetization of (a) sample means from the road section, and (b) successive disks from a core through sample C16. Symbols are as in fig. 2, and the regions enclosed by the dashed lines are those occupied by the normal (N) or reversed (R) disks. The remainder have intermediate directions.

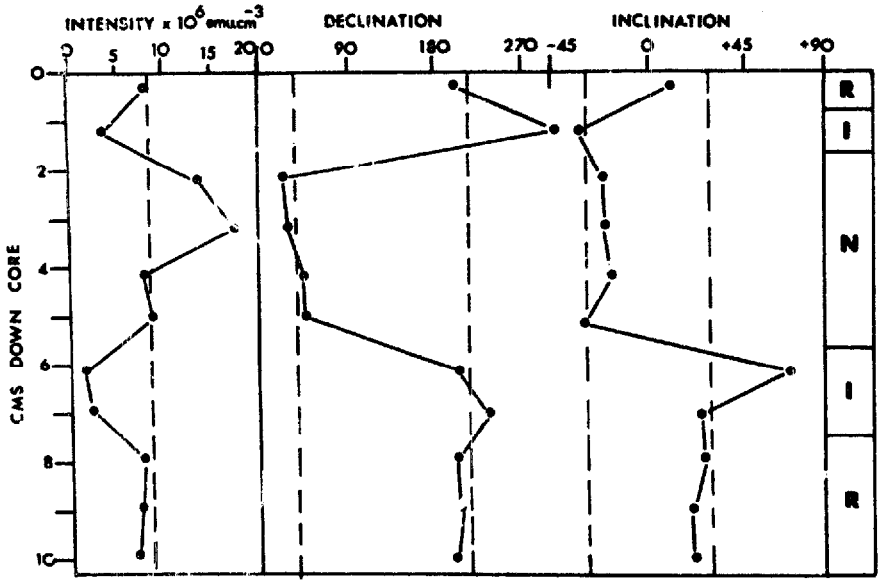


Fig. 5. Magnetic stratigraphy of the 10 cm core through sample C16. The normal (N), reversed (R) and intermediate (I) zones indicated on the right.

intermediate directions and four normally magnetized ones. Near the top there is another intermediate direction and the disk at the top of the core is again reversely magnetized. The intensities of magnetization of the intermediate disks (fig. 5) are only about one quarter of the values of the normal or reversed disks. This is consistent with the view that the residual field strength during a geomagnetic reversal is only one quarter or one fifth of its normal value [20,21].

At the center of the core the two disks which record intermediate directions cover a thickness of sediment of about 1 cm. Estimates of the time taken for the field to reverse itself vary from 10^3 to 10^4 yr [22,23]. The rate of sedimentation can thus be calculated as lying between 0.1 and 1 cm per thousand years, and the length of the whole core therefore represents a time span of between 10^4 and 10^5 y. Thus there appears to be every justification in regarding each sample as a separate point in time; indeed, there is a strong case for suggesting that each sample mean itself averages out secular variation. Since the 10 samples investigated in this section cover a stratigraphic thickness of about 30 m, there is a time span of between 3 and 30 my represented. The mean of 11 vectors in table 2 and fig. 4a, and the pole position calculated from it almost certainly represent good estimates of the axial dipole field direction and palaeomagnetic pole position in Lower Cambrian times with respect to the Indian sub-continent.

The palaeolatitude of 17° is consistent with the hot dry arid conditions which were thought to have prevailed during the deposition of these rocks [14].

6. India and Gondwanaland

The new Lower Cambrian results from the Indian sub-continent make it possible to fit the final pieces of the jig-saw puzzle of Gondwanaland into place. I have previously shown that South America and Africa can be located with respect to each other by comparing their polar paths for most of the Palaeozoic [5] Australia can be located with respect to these two continents by comparing the Upper Palaeozoic paths [6], since no Lower Palaeozoic data of any consequence has yet been reported from Australia. The resulting arrangement of these three continents and the common path so deduced is shown in fig. 6. Approximate ages are indicated at points along the common path.

To fit India to the configuration outlined above

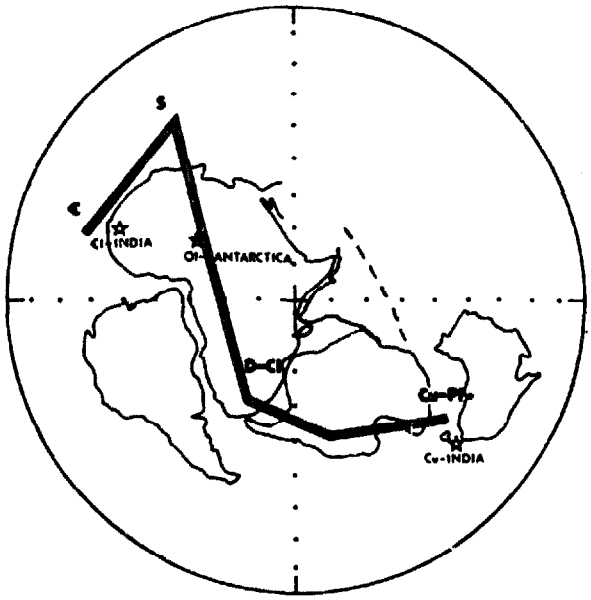


Fig. 6. Reconstruction of Gondwanaland with respect to the present co-ordinates of Africa. The solid line is the common polar path for the Palaeozoic inferred by McElhinny [5] for South America and Africa, and also for Australia during the Upper Palaeozoic, when these three continents are placed as shown. Approximate ages are indicated at various points (heavy lettering). The position of India is determined by making its Lower Cambrian pole (this paper) and Upper Carboniferous pole [12] coincide with the beginning and end of the common path respectively. With Antarctica in the position shown, its Lower Ordovician pole [24] lies close to the Lower Palaeozoic portion of the common polar path.

requires that this continent have two points (palaeomagnetic poles) defined for comparison with the common path. The new Lower Cambrian pole deduced in this paper provides one point and a recent result from the Upper Carboniferous Talchir Series [12], provides another point towards the end of the Palaeozoic. This places India in the position shown in fig. 6 adjacent to East Africa. In this diagram Africa is located in its present position and each of the other continents have been moved so that their poles lie on the common path defined with respect to Africa in its present position

Antarctica can only be located by using a Lower Ordovician pole [24]. As yet only one point is available for Antarctica for the Palaeozoic, but its location is effectively restrained by the relative positions of the other four continents already determined. The resulting configuration of Gondwanaland is thus determined as in fig. 6. This arrangement of the continents is virtually identical with that first proposed by du Toit [25], and it shows in particular that India and Australia are not adjacent. The geochronological evidence in favour of this [9] may thus be either coincidental or, more likely, not definitive enough. The interpretation of marine magnetic anomalies in the Indian Ocean in terms of the sea-floor spreading hypothesis [26,27] appears to favour the juxtaposition of India and East Africa as suggested here.

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